# REB, The Book
# Example 6.6.5 Solution

A 10 gal CSTR with a 1.25 gal shell is going to be used for the conversion of A and B according to reactions (1), (2), and (3). Both the reactor and the shell are perfectly mixed. Before flow into the reactor begins, it is filled with a solution containing only B at a concentration of 3 mol $gal^{-1}$ and a temperature of 20 °C, while the shell contains cooling water at 20 °C. To start up the reactor, 0.5 gal $min^{-1}$ of a feed consisting of a liquid solution of A (7.5 mol $gal^{-1}$) and B (3 mol $gal^{-1}$) at 50 °C starts flowing into the reactor and simultaneously cooling water at 20 °C starts flowing into the shell at a rate of 250 g $min^{-1}$. The heat transfer coefficient between the reactor and the shell is 190 cal $ft^{-2}$ $min^{-1}$ $K^{-1}$ and the heat transfer area is 4 $ft^2$. The entire process operates at a constant pressure of 1 atm.

$$A + B \rightarrow W + Z \quad (1)$$

$$A + B \rightarrow X + Z \quad (2)$$

$$A + B \rightarrow Y + Z \quad (3)$$

Each of the reactions is first order in the concentration of A and first order in the concentration of B. The pre-exponential factors, activation energies and heats of reaction are listed in the table below. The heats of reaction may be taken to be constant. The cooling water has a constant density of 1 g $cm^{-3}$ and a constant heat capacity of 1 cal $g^{-1}$ $K^{-1}$. The reacting fluid may be assumed to have a constant heat capacity of 1600 cal $gal^{-1}$ $K^{-1}$.

Plot the temperature of the reacting fluid, the temperature of the cooling water in the jacket, and the concentration of B in the reactor during the first 30 minutes of the start-up process.

Table 1. Reaction Data

| Reaction | $k_0$ (gal $mol^{-1}$ $min^{-1}$) | E (kcal $mol^{-1}$) | ΔH (kcal $mol^{-1}$) |
|---|---|---|---|
| 1 | $4.3 \times 10^8$ | 14.2 | -11.0 |
| 2 | $2.7 \times 10^8$ | 16.1 | -11.6 |
| 3 | $3.0 \times 10^8$ | 14.8 | -12.1 |

## *Assignment Summary*

Reactions

$$A + B \rightarrow W + Z \quad (1)$$

$$A + B \rightarrow X + Z \quad (2)$$

$$A + B \rightarrow Y + Z \quad (3)$$

Rate Expressions

$$r_1 = k_{1,0} \exp\left(\frac{-E_1}{RT}\right) C_A C_B \quad (4)$$

$$r_2 = k_{0,2} \exp\left(\frac{-E_2}{RT}\right) C_A C_B \quad (5)$$

$$r_3 = k_{0,3} \exp\left(\frac{-E_3}{RT}\right) C_A C_B \quad (6)$$

Reactor: Non-isothermal, transient CSTR

Given Constants: $V$= 10 gal, $V_{ex}$ = 1.25 gal, $C_{B,0}$ = 3 mol gal$^{-1}$, $T_0$ = (20 + 273.15) K, $\dot{V}_{in}$ = 0.5 gal min$^{-1}$, $C_{A,in}$ = 7.5 mol gal$^{-1}$, $C_{B,in}$ = 3 mol gal$^{-1}$, $T_{in}$ = (50 + 273.15) K, $T_{ex,0}$ = $T_{ex,in}$ = (20 + 273.15) K, $\dot{m}_{ex,in}$ = 250 g min$^{-1}$, $U$ = 190 cal ft$^{-2}$ min$^{-1}$ K$^{-1}$, $A$ = 4 ft$^2$, $P$ = 1 atm, $\rho_{ex}$ = 1 g cm$^{-3}$, $\tilde{C}_{p,ex}$ = 1 cal g$^{-1}$ K$^{-1}$, $\breve{C}_p$ = 1600 cal gal$^{-1}$ K$^{-1}$, $k_{0,j}$ in Table 1, $E_j$ in Table 1, $\Delta H_j$ in Table 1, and $t_f$ = 30 min.

Deliverables: $T$ *vs.* $t$, $T_{ex}$ *vs.* $t$, and $C_B$ *vs.* $t$ for $0 < t < t_f$

## *Completion Strategy*

This assignment asks for the response of a transient CSTR. The reacting fluid temperature and exchange fluid temperature over time can be found directly by solving the transient CSTR design equations. Solving the design equations will also yield the

outlet molar flow rate of reagent B over time, from which the outlet concentration of B over time can be calculated.

### *Formulation of the Calculations*

**CSTR Model Function**

CSTR Design Equations:

$$\frac{d\dot{n}_A}{dt} = \frac{\dot{V}}{V}\left(\dot{n}_{A,in} - \dot{n}_A - V\left(r_1 + r_2 + r_3\right)\right) \tag{7}$$

$$\frac{d\dot{n}_B}{dt} = \frac{\dot{V}}{V}\left(\dot{n}_{B,in} - \dot{n}_B - V\left(r_1 + r_2 + r_3\right)\right) \tag{8}$$

$$\frac{d\dot{n}_W}{dt} = \frac{\dot{V}}{V}\left(-\dot{n}_W + Vr_1\right) \tag{9}$$

$$\frac{d\dot{n}_X}{dt} = \frac{\dot{V}}{V}\left(-\dot{n}_X + Vr_2\right) \tag{10}$$

$$\frac{d\dot{n}_Y}{dt} = \frac{\dot{V}}{V}\left(-\dot{n}_Y + Vr_3\right) \tag{11}$$

$$\frac{d\dot{n}_Z}{dt} = \frac{\dot{V}}{V}\left(-\dot{n}_Z + V\left(r_1 + r_2 + r_3\right)\right) \tag{12}$$

$$\frac{dT}{dt} = \frac{\dot{Q} - \dot{V}_{in}\breve{C}_p\left(T - T_{in}\right) - V\left(r_1\Delta H_1 + r_2\Delta H_2 + r_3\Delta H_3\right)}{V\breve{C}_p} \tag{13}$$

$$\frac{dT_{ex}}{dt} = \frac{-\dot{Q} - \dot{m}_{ex}\tilde{C}_{p,ex}\left(T_{ex} - T_{ex,in}\right)}{\rho_{ex}V_{ex}\tilde{C}_{p,ex}} \tag{14}$$

CSTR Model Variables: $\underline{t}$, $\underline{\dot{n}_A}$, $\underline{\dot{n}_B}$, $\underline{\dot{n}_W}$, $\underline{\dot{n}_X}$, $\underline{\dot{n}_Y}$, $\underline{\dot{n}_Z}$, $\underline{T}$, and $\underline{T_{ex}}$

Additional Computable Unknowns: $\dot{n}_{A,in}$, $r_1$,$r_2$, $r_3$, $C_A$, $C_B$, $\dot{V}$, $\dot{n}_{B,in}$, and $\dot{Q}$

$$\dot{n}_{i,in} = \dot{V}_{in} C_{i,in} \qquad i = A \text{ and } B \tag{15}$$

$$C_i = \frac{\dot{n}_i}{\dot{V}} \qquad i = A \text{ and } B \tag{16}$$

$$\dot{V} = \dot{V}_{in} \tag{17}$$

$$\dot{Q} = UA\left(T_{ex} - T\right) \tag{18}$$

IVODE Solver Requirements:

1. Initial values of the CSTR model variables

$$t\Big|_{t=0} = 0 \tag{19}$$

$$\dot{n}_i\Big|_{t=0} = 0 \qquad i = A, W, X, Y, \text{ and } Z \tag{20}$$

$$\dot{n}_B\Big|_{t=0} = C_{B,0}\dot{V} \tag{21}$$

$$T\Big|_{t=0} = T_0 \tag{22}$$

$$T_{ex}\Big|_{t=0} = T_{ex,0} \tag{23}$$

2. Stopping criterion

$$t = t_f \tag{24}$$

3. Derivatives function that
   a. receives the CSTR model variables at the start of an integration step
   b. calculates the additional unknowns using equations (4) - (6), and (15) - (18)
   c. evaluates and returns the design equation derivatives using equations (7) - (14)

**Deliverables Function**

1. Call the CSTR model function to get the CSTR model variables
2. Calculate the outlet concentration of B over time

$$\underline{C}_B = \frac{\dot{n}_B}{\dot{V}} \tag{25}$$

### *Implementation of the Calculations*

The Python script, example_6_6_5.py, and Matlab function, example_6_6_5.m, that were written to perform the calculations accompany this solution.

### *Results and Discussion*

The startup begins with both the reactor and the shell at 20 °C, and with the reactor containing only reagent B. The reactor feed flow that is then started is at 50 °C and contains both A and B. All three reactions are exothermic, so the temperature of the reacting fluid is expected to increase both due to the warmer feed entering it and due to the heat released by the reactions. As heat is transferred to the exchange fluid, its temperature will also rise. The concentration of B in the reactor is expected to decrease over time as well. This is due in part to the lower concentration of B in the feed and in part to the consumption of B in the reactions. Since no other changes are made after starting the feed and exchange fluid feeds, the system is expected to approach a steady state where all of the outlet variables are constant.

This behavior is confirmed in Figures 1 through 3 which show the reacting fluid temperature, exchange fluid temperature and concentration of B during the first 30 min of startup. The temperatures both increase and the concentration of B decreases over time. All three quantities appear to be approaching steady-state values.

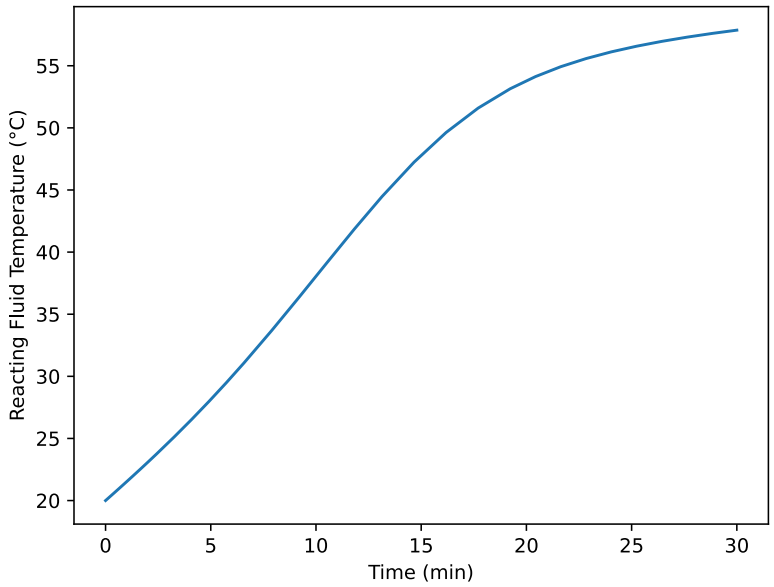



*Figure 1. Reacting fluid temperature during the first 30 minutes of startup.*

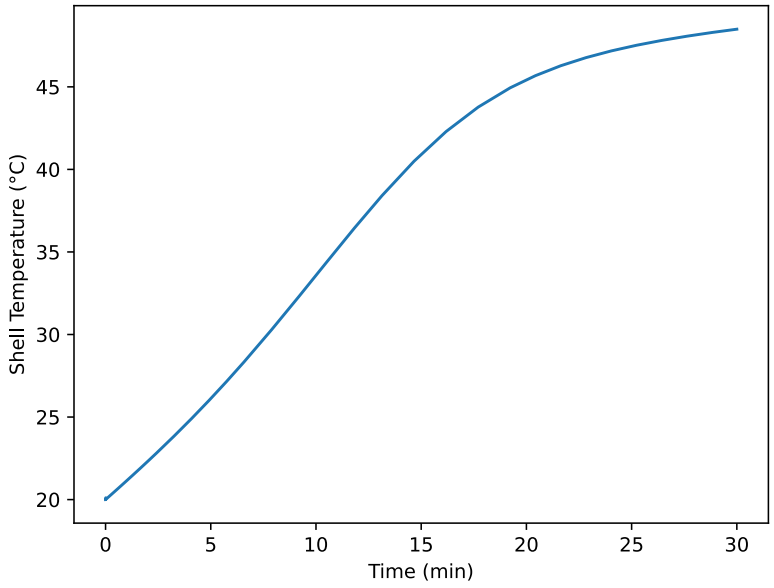



*Figure 2. Exchange fluid temperature during the first 30 minutes of startup.*

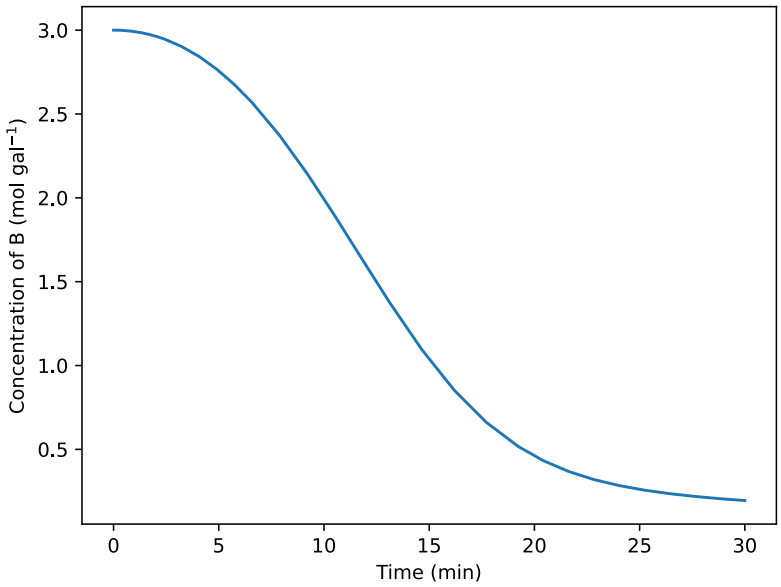



*Figure 3. Concentration of reagent B during the first 30 minutes of startup.*

### *Summary*

As shown in Figure 1, the reacting fluid temperature rises from 20 °C to ca. 57 °C during the first 30 minutes of startup. Figure 2 shows that during the same period, the exchange fluid temperature increases from 20 °C to ~48 °C, and Figure 3 shows that the concentration of B drops from 3 mol gal⁻¹ to approximately 0.2 mol gal⁻¹. These variables are still changing after 30 minutes, but they appear to be approaching steady-state values.